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Installation

Step1) Download Oracle Database software

From [Database Software Downloads | Oracle](https://www.oracle.com/database/technologies/xe-downloads.html)

Click the highlighted option below to download the software (Currently latest version is 21c, always prefer the latest one)

Oracle Database XE Downloads

Oracle Database 21c Express Edition

Download	Description
Oracle Database 21c Express Edition for Windows x64	(1,967,615,483 bytes - October 08, 2021) [Sha256sum: 939742c3305c466566a55f607638621b6aa7033a183175f6bcd6cffb48e6bc3f]
Oracle Database 21c Express Edition for Linux x64 (OL8)	(2,339,651,768 bytes - September 08, 2021) [Sha256sum: f8357b432de33478549a76557e8c5220ec243710ed86115c65b0c2bc00a848db]
Oracle Database 21c Express Edition for Linux x64 (OL7)	(2,339,017,432 bytes - September 08, 2021) [Sha256sum: 4c8f40a19d4d1a2f00e46df022943a04cc13fe62aed27c4c66a137e72f513c36]

Why is Oracle Database software required?

Oracle Database software provides the database engine, which is the brain that stores data and understands and executes SQL commands. Without this database engine, we cannot work with SQL.

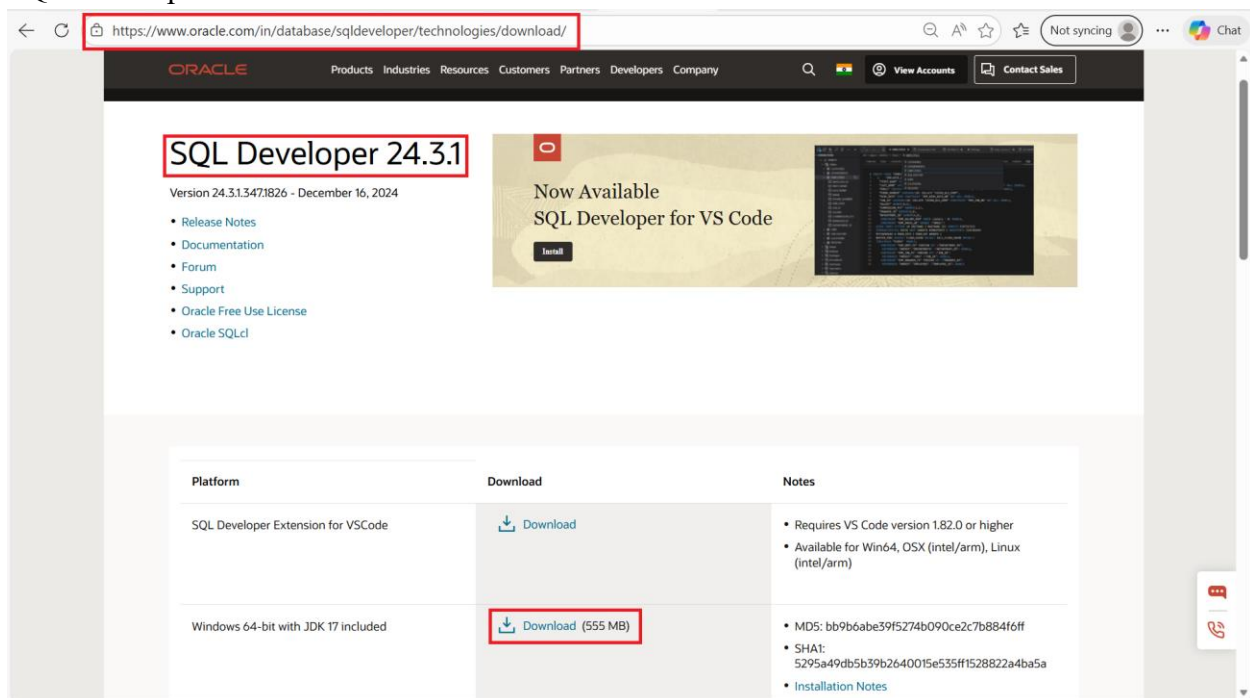
Step2) Download SQL Developer Tool

From [Oracle SQL Developer Downloads](https://www.oracle.com/in/database/sqldeveloper/technologies/download/) | [Oracle India](#)

Click the highlighted Download option below to download the software

Note – In the below screenshot it is saying “with Java”, always download SQL Developer Tool which is included with Java because the entire software is written in java.

SQL Developer is a Java-based UI tool.



The screenshot shows the Oracle SQL Developer download page. The URL in the browser is <https://www.oracle.com/in/database/sqldeveloper/technologies/download/>. The page features the Oracle logo and navigation links. The main heading is "SQL Developer 24.3.1" with the version number highlighted in a red box. Below this, there are links for Release Notes, Documentation, Forum, Support, Oracle Free Use License, and Oracle SQLcl. A banner for "Now Available SQL Developer for VS Code" is also present. The download section is a table with three columns: Platform, Download, and Notes.

Platform	Download	Notes
SQL Developer Extension for VSCode	Download	- Requires VS Code version 1.82.0 or higher - Available for Win64, OSX (intel/arm), Linux (intel/arm)
Windows 64-bit with JDK 17 included	Download (555 MB)	- MD5: bb9b6abe39f5274b090ce2c7b884f6f - SHA1: 5295a49db5b39b2640015e535ff1528822a4ba5a Installation Notes

Why is SQL Developer Tool required?

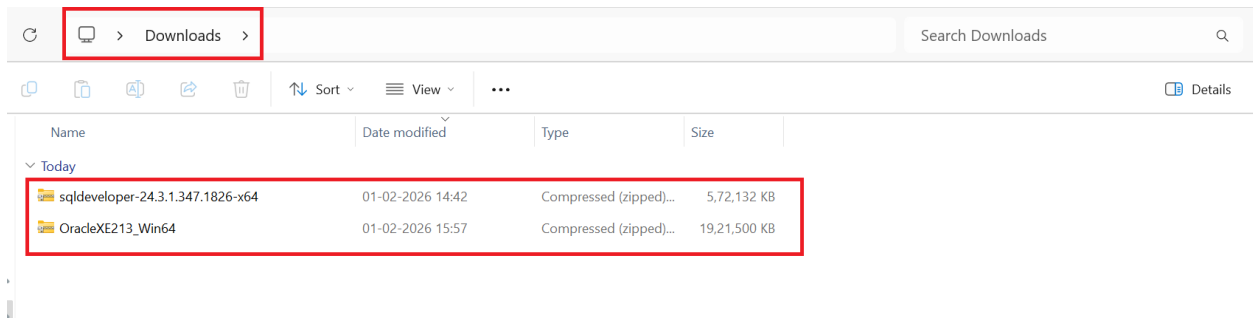
SQL Developer is a tool that helps us write and run SQL and PL/SQL programs easily. But it cannot run the code by itself without Oracle Database engine.

Since we downloaded and installed the Oracle Database engine in Step1, the SQL Developer Tool connects to it, to execute our code and manage the data.

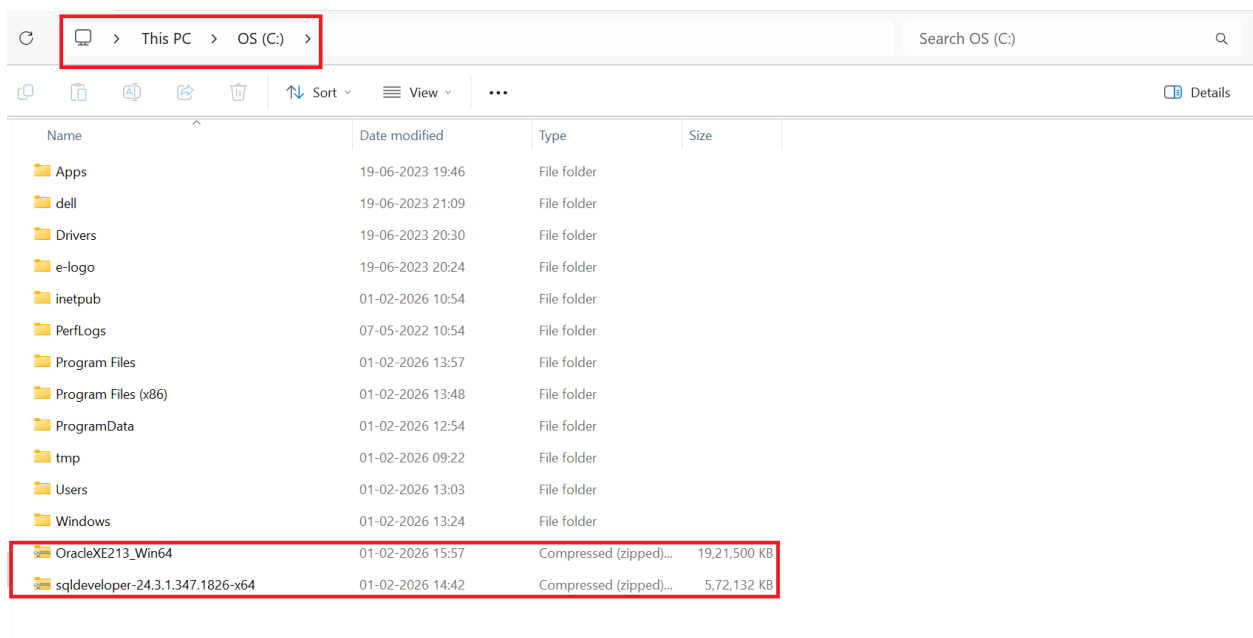
Step3) Place the downloaded zipped folder under C drive and extract it

This is to ensure smooth connection between SQL Developer Tool and the Oracle Database engine as SQL Developer Tool requires database engine internally

Cut From Downloads -



Paste in C -

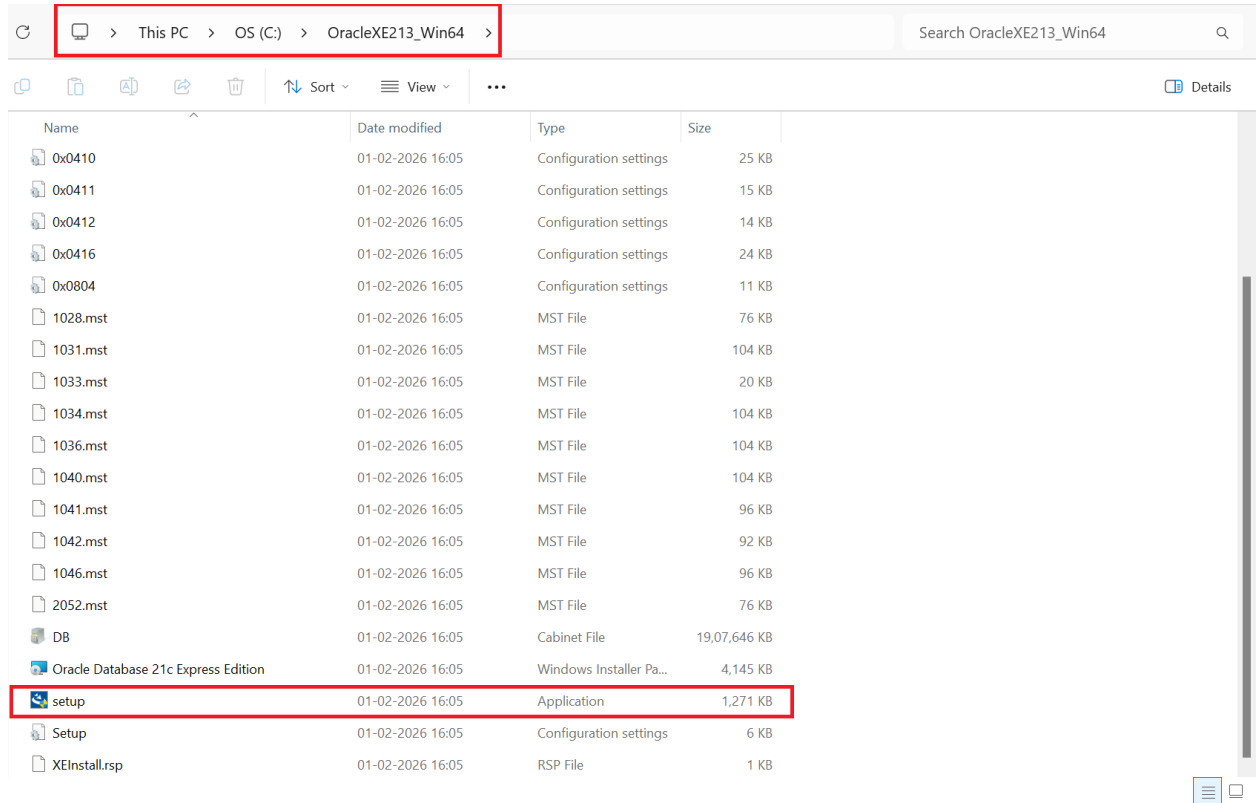


Now, extract both folders.

Step4) Install Oracle Database Engine and Software Developer Tool)

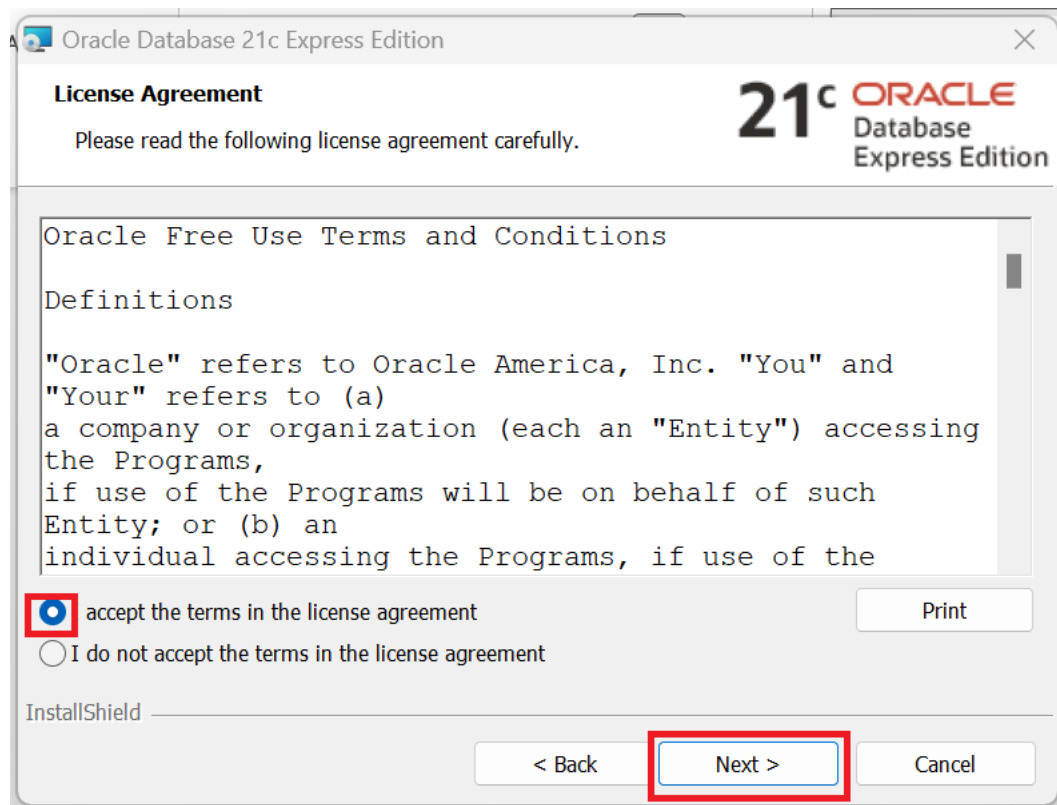
Install Oracle Database Engine

→ Open Oracle Database folder and double click on setup application file



→ Click on **Next**

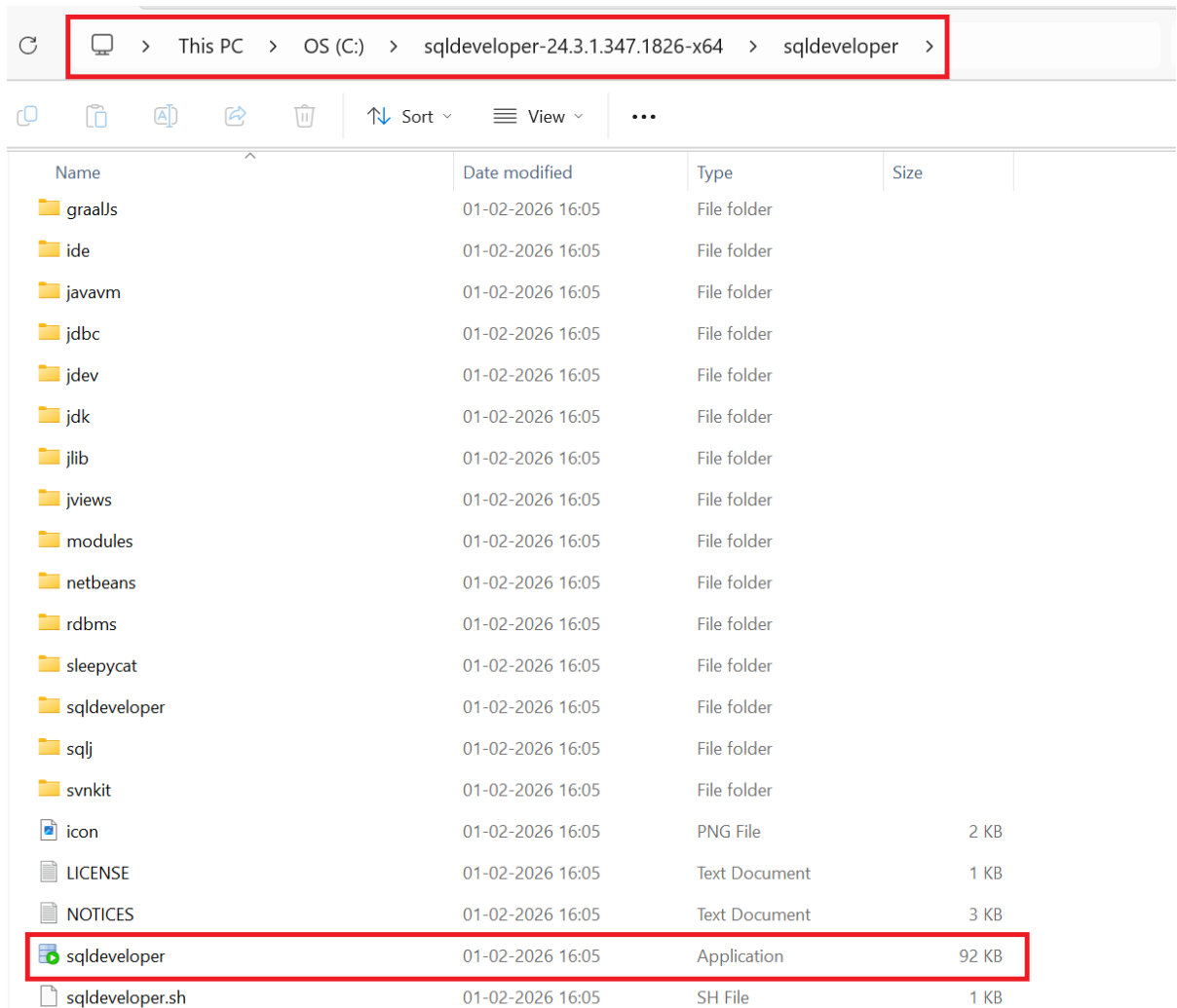
→ Select accept radio button and click on **Next**



- Save the destination folder location and click **Next**
- **Set easy password** and remember the password
- Take the screenshot of wizard which includes Destination Folder, Oracle Home, Oracle Base and click **Install**

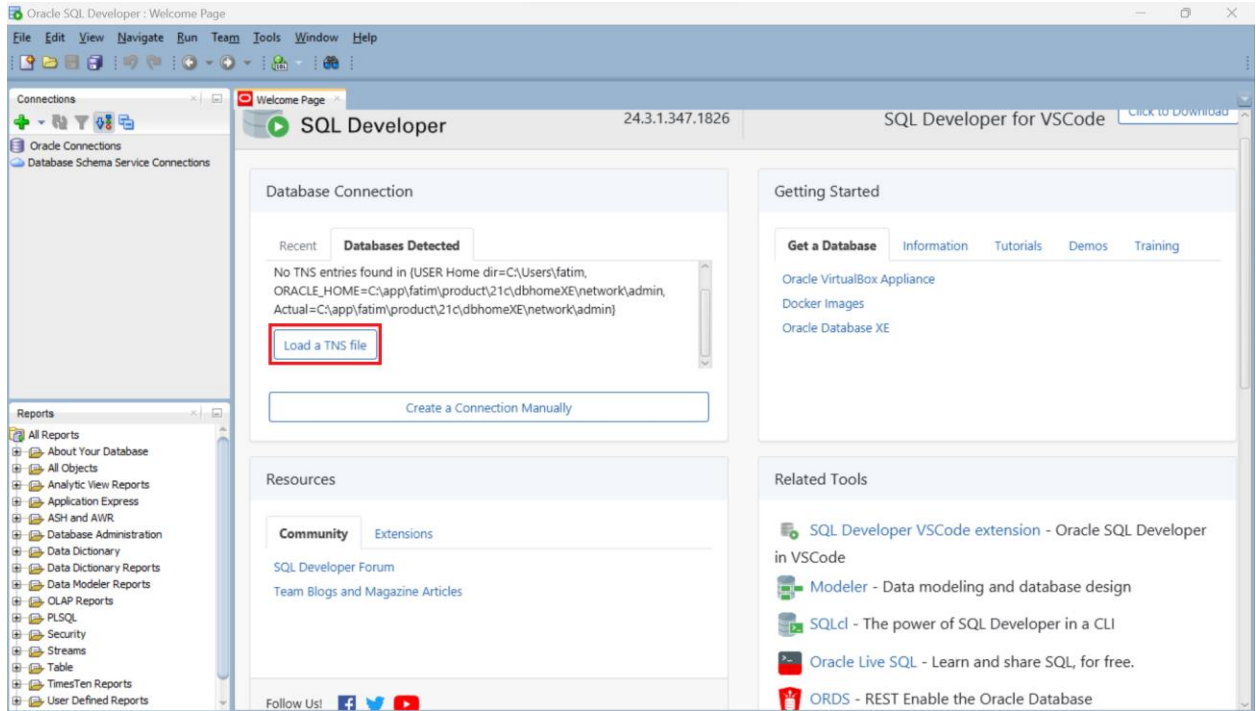
Install Software Developer Tool

→ Open SQL Developer folder and double click on sqldeveloper application file and install it

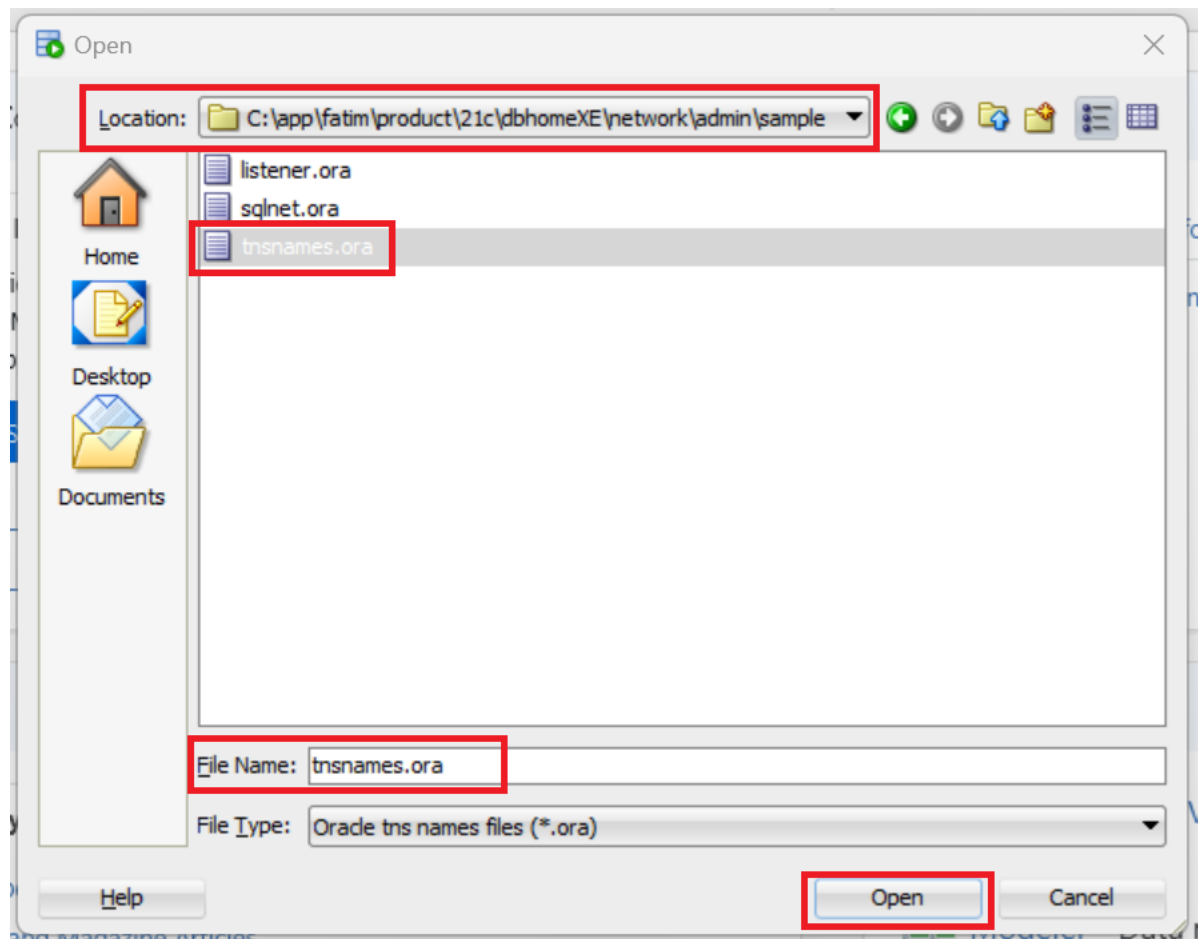


Step5) Connect SQL Developer Tool with Database Engine

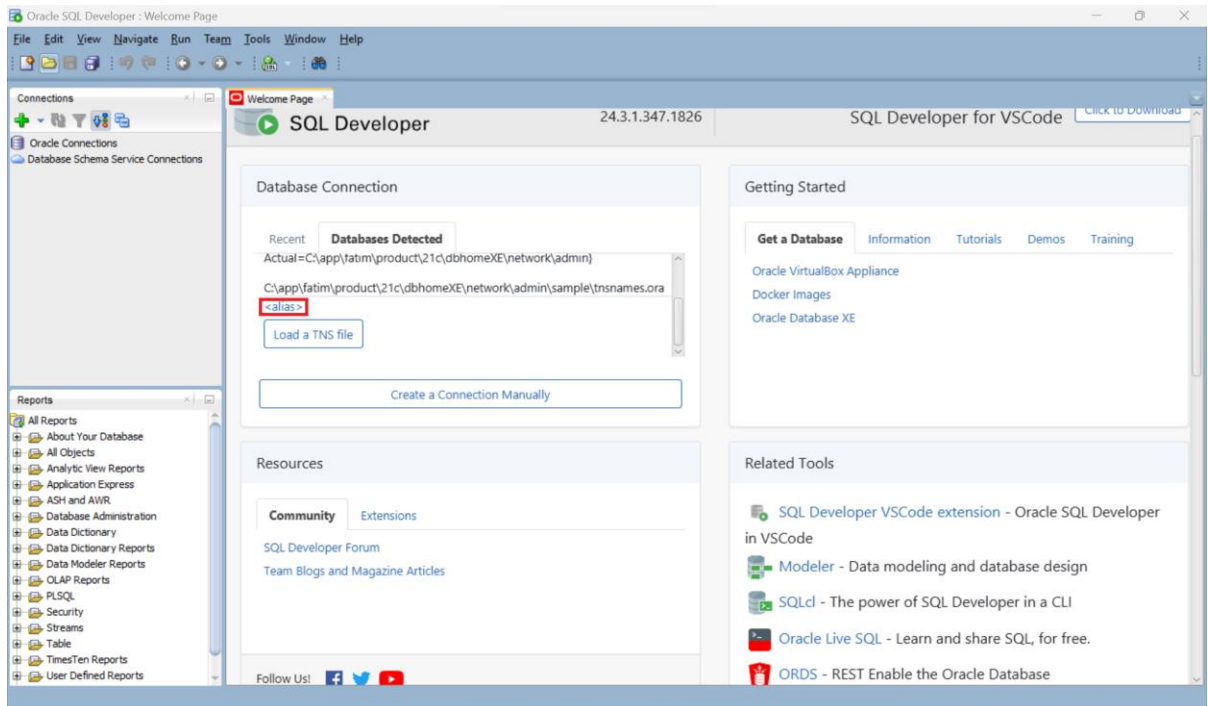
→ Click on Load a TNS file



→ Select tnsnames.ora file



→ After opening the above file, we may encounter a problem that is “Instead of database name it may shows up “<alias>” -



→ Solution for the above problem is as follows -

1. Go to Google and type sample tnsnames.ora file

The screenshot shows a Google search result for the query "tnsnames ora oracle xe sample". The search bar at the top contains the query. Below the search bar, a list of results is displayed. The first result is titled "Connecting to the Root Container Database (CDB) XE" and describes a service that connects to the main container. Below this result, a code block is shown with the title "ora" and a copy icon. The code block contains the following text:

```
XE =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP)(HOST = localhost)(PORT = 1521))
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = XE)
    )
  )
```

To the right of the code block, there are three search results from Oracle Help Center, each titled "6 Connecting to Oracle Database X" and providing information about local naming parameters and service names.

2. Query Ip address of machine – open command prompt, hit ipconfig and copy IPV4 address

```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows [Version 10.0.22631.6199]
(c) Microsoft Corporation. All rights reserved.

C:\Users\fatim>ipconfig

Windows IP Configuration

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

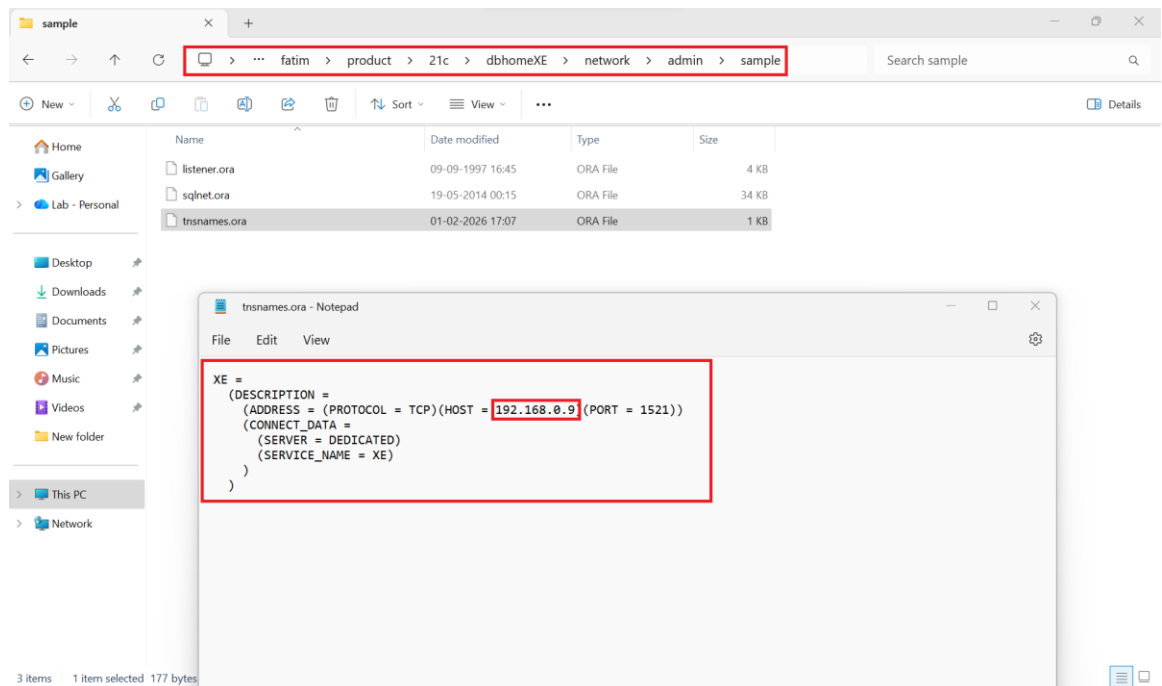
    Connection-specific DNS Suffix  . : domain.name
    Link-local IPv6 Address . . . . . : fe80::bae2:d3e9:478f:5374%16
    IPv4 Address. . . . . : 192.168.0.9
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 192.168.0.1

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :

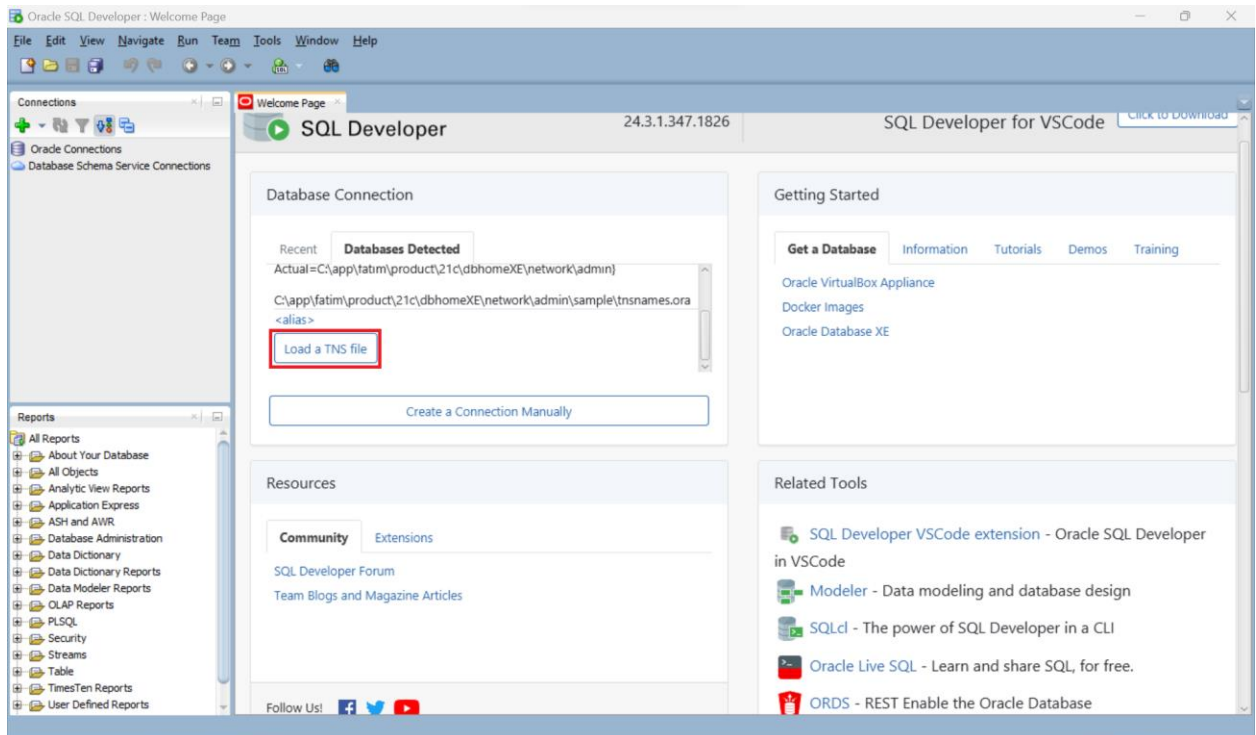
C:\Users\fatim>
```

3. Copy the connection description from (1) and IPV4 address from (2) and paste it in tnsnames.ora file -

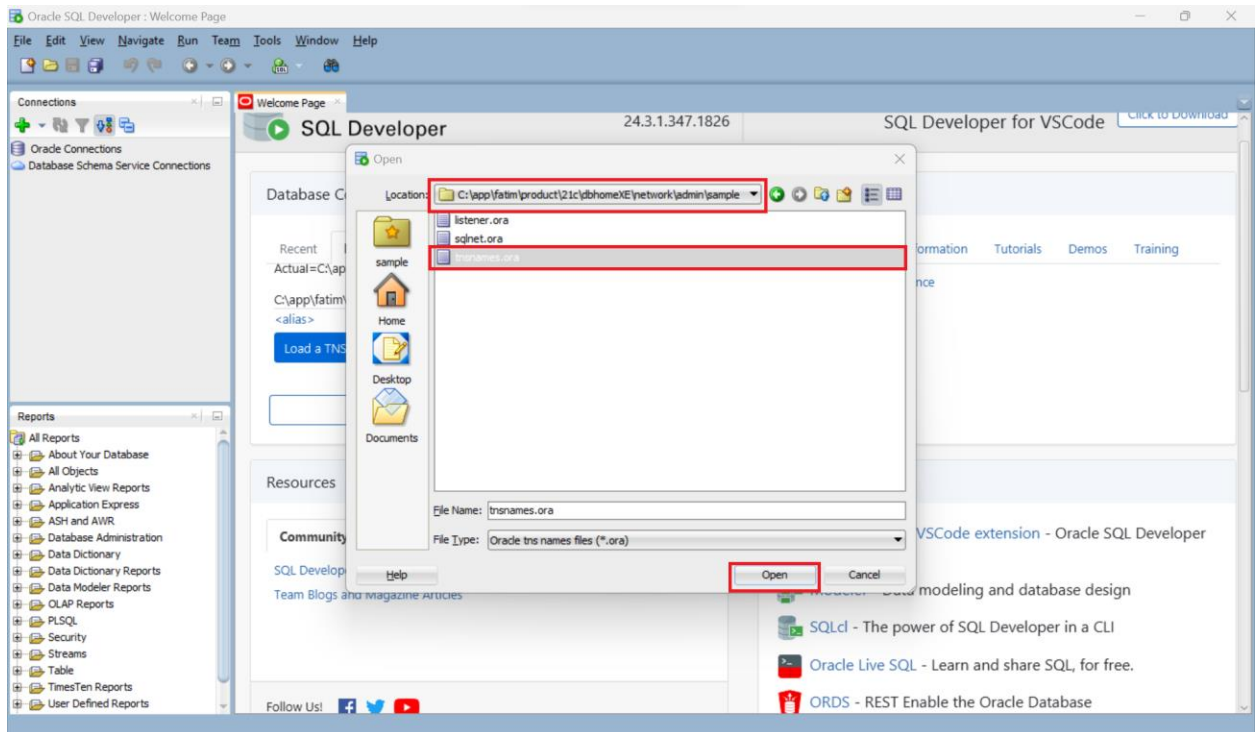


4. Save and Close it

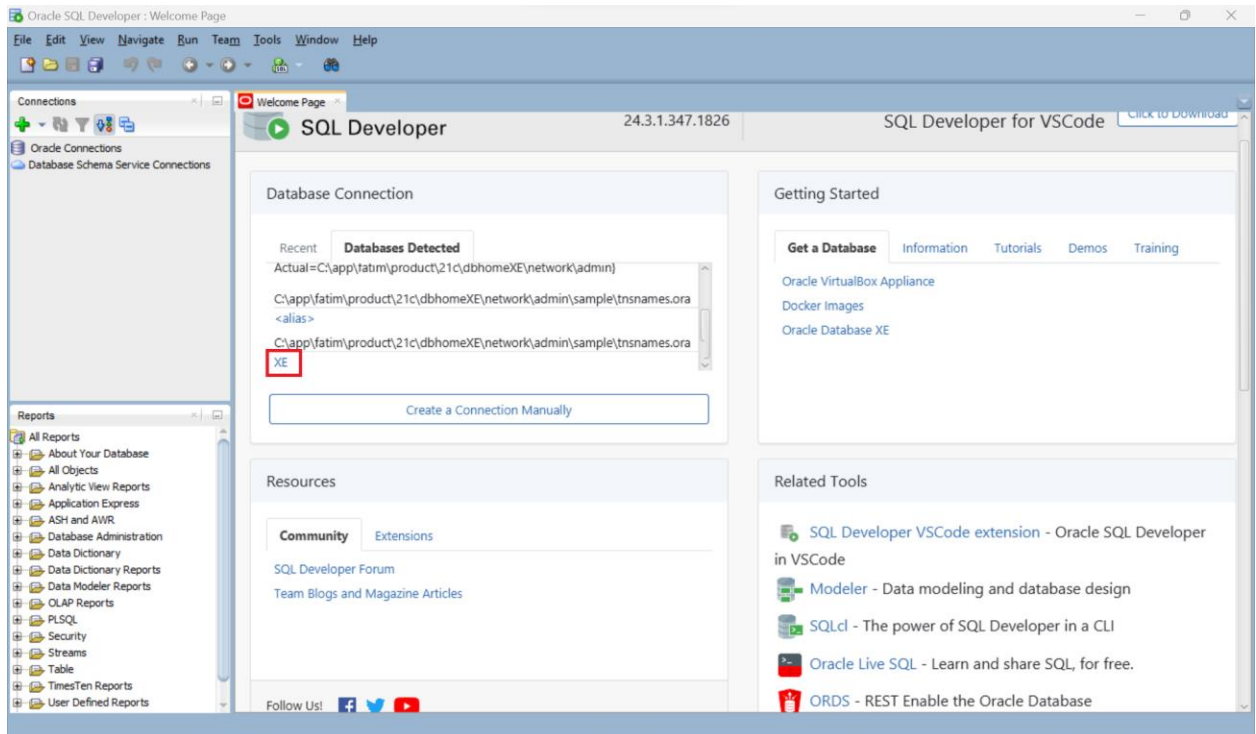
→ Now open SQL Developer Tool and try again to Load a TNS file –



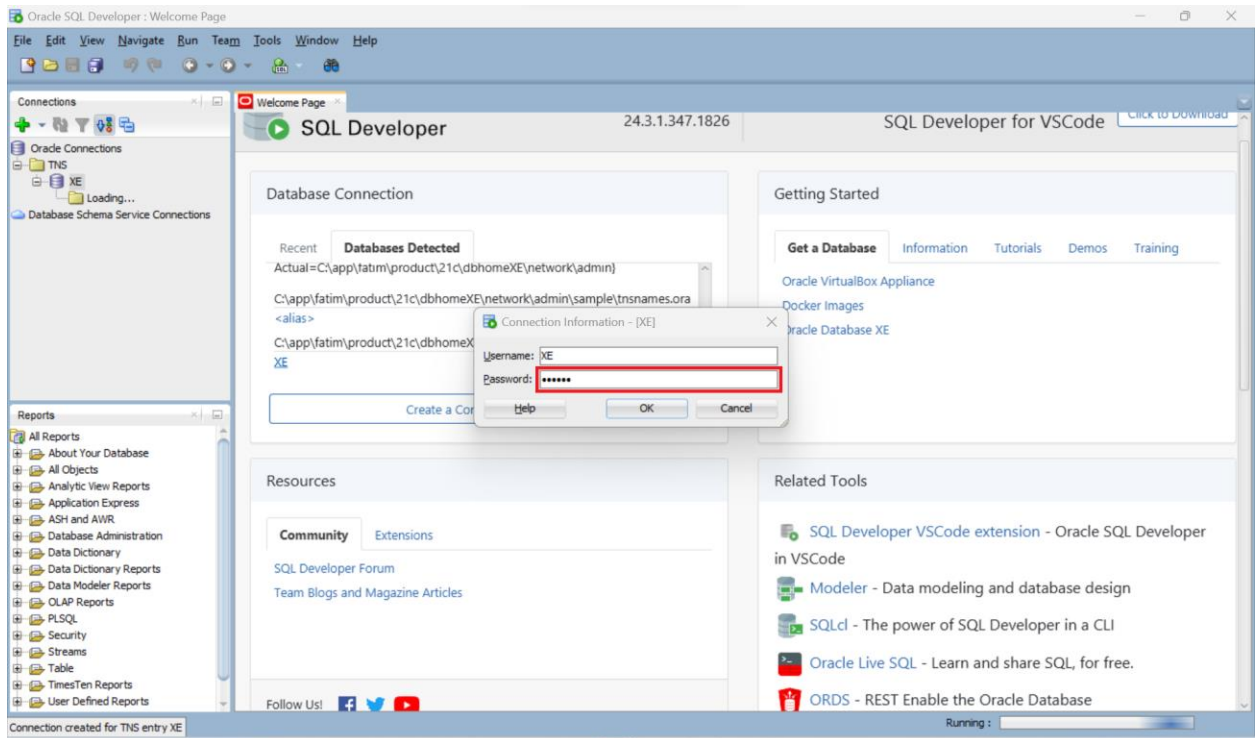
→ Select tnsnames.ora and click Open



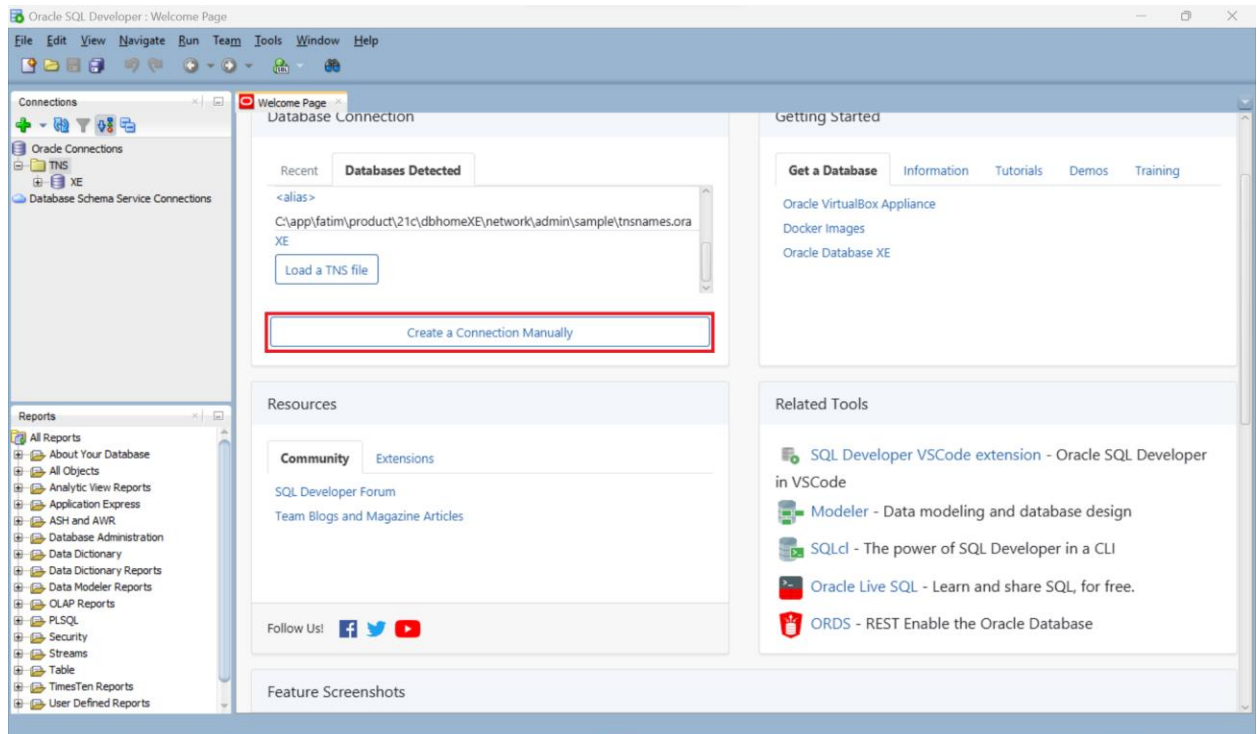
→ Click on XE database



→ Enter the password which was set during Oracle Database Engine Software installation and click on OK –



→ After entering the password, sometimes we may encounter invalid credentials issue, at this step we can click on “Create a Connection Manually”



→ Click on the connection name XE in the left panel.

In the center panel, enter the following:

Username: system

Password: The password was set during Oracle Database installation.

Click the Test button.

Check the status: it should show Success.

The screenshot shows the 'New / Select Database Connection' dialog box. It has a table on the left with two columns: 'Connection Name' and 'Connection Details'. The first row is highlighted with a red box and contains 'XE' and 'XE@//192.168.0...'. Below the table, the status 'Status : Success' is displayed in a red box. To the right of the table, there are configuration fields for a new connection. The 'Name' field is 'XE'. The 'Database Type' is 'Oracle'. The 'User Info' tab is selected, showing 'Authentication Type' as 'Default', 'Username' as 'system', and 'Password' as '*****'. The 'Role' is 'default' and 'Save Password' is unchecked. The 'Connection Type' is 'Basic'. The 'Details' tab is selected, showing 'Hostname' as '192.168.0.9', 'Port' as '1521', and 'Service name' as 'XE'. The 'Test' button at the bottom is highlighted with a red box.

Connection Name	Connection Details
XE	XE@//192.168.0...

Name: XE

Database Type: Oracle

User Info

Authentication Type: Default

Username: system

Password: *****

Role: default

Save Password: ☐

Connection Type: Basic

Details

Advanced

Hostname: 192.168.0.9

Port: 1521

☐ SID

☒ Service name: XE

Status : Success

Help Save Clear Test Connect Cancel

→ Click on Connect

New / Select Database Connection

Connection Name	Connection Details
XE	XE@//192.168.0....

Name: XE Color

Database Type: Oracle

User Info Proxy User

Authentication Type: Default

Username: system Role: default

Password: Save Password

Connection Type: Basic

Details Advanced

Hostname: 192.168.0.9

Port: 1521

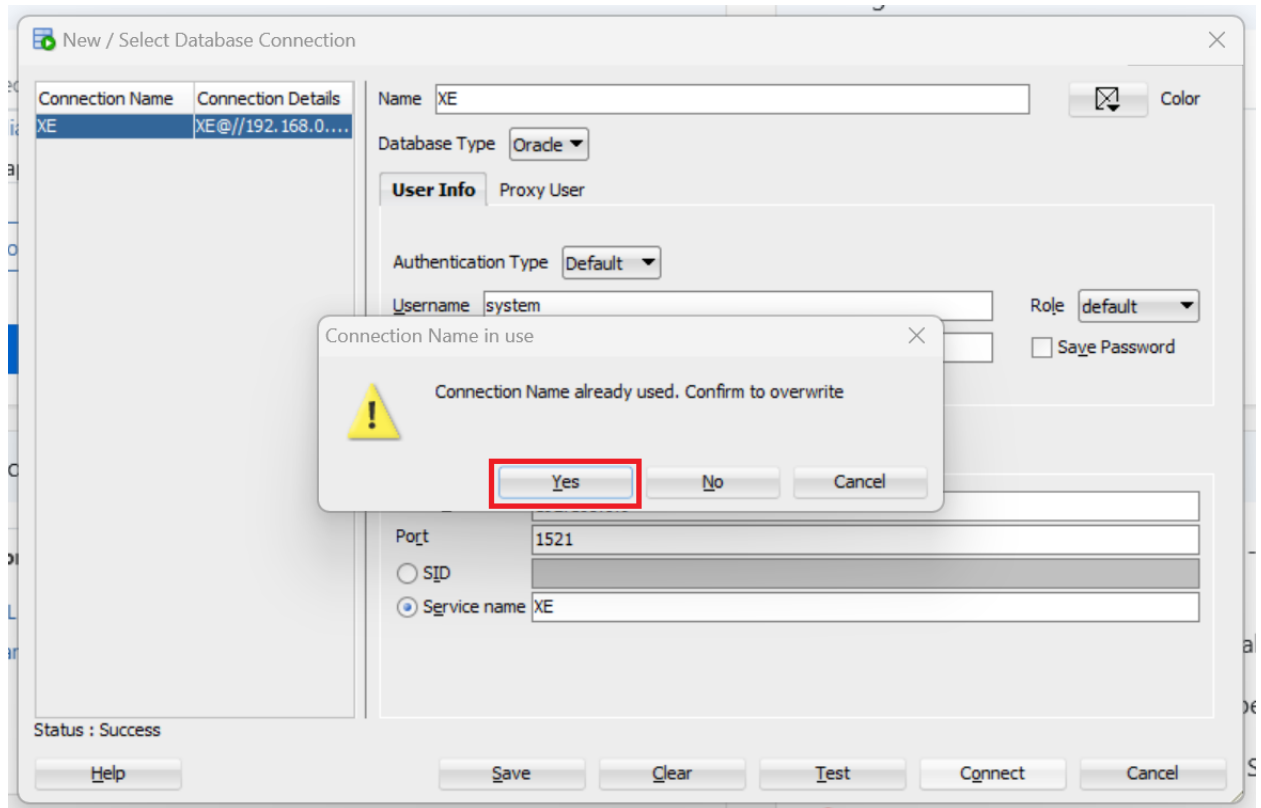
☐ SID

☒ Service name: XE

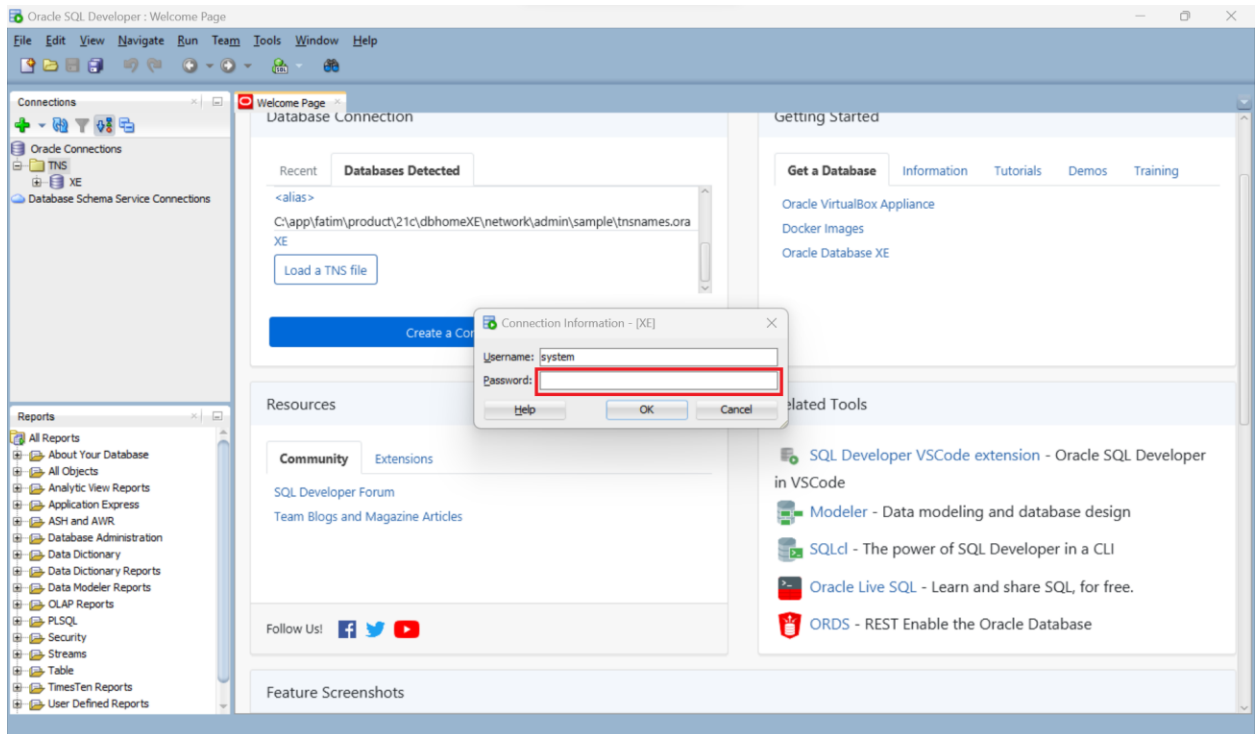
Status: Success

Help Save Clear Test Connect Cancel

→ If it shows below confirmation screen, then click on Yes

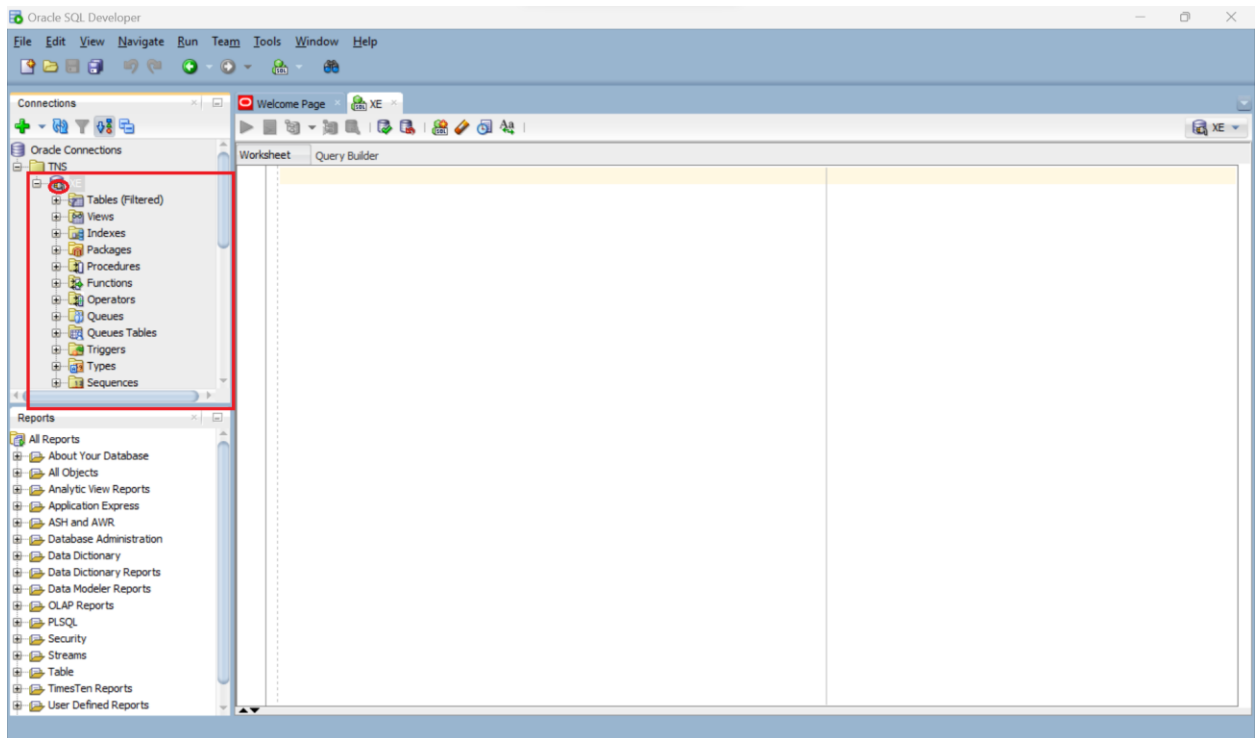


→ Enter the password you set during Oracle Database installation and click on OK



- SQL Developer Tool is successfully connected with Oracle Database Software -
- ✓ Small Plug icon appears next to the connection which ensures it is connected successfully

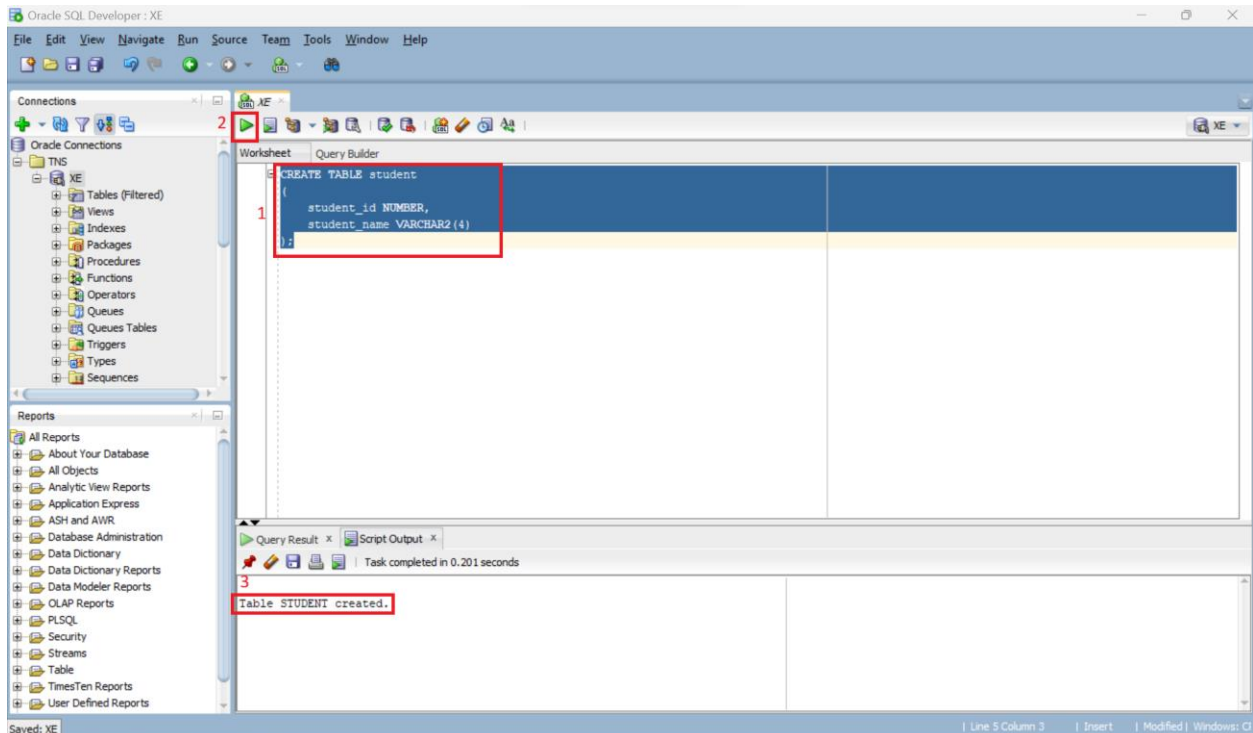
✓ Also, if we expand XE we can see Tables, Views, Indexes and so on



Writing First SQL Query

Step1) Create a table

- 1) Write the code
- 2) Click on Run button
- 3) Output will be displayed “Table tableName created.”



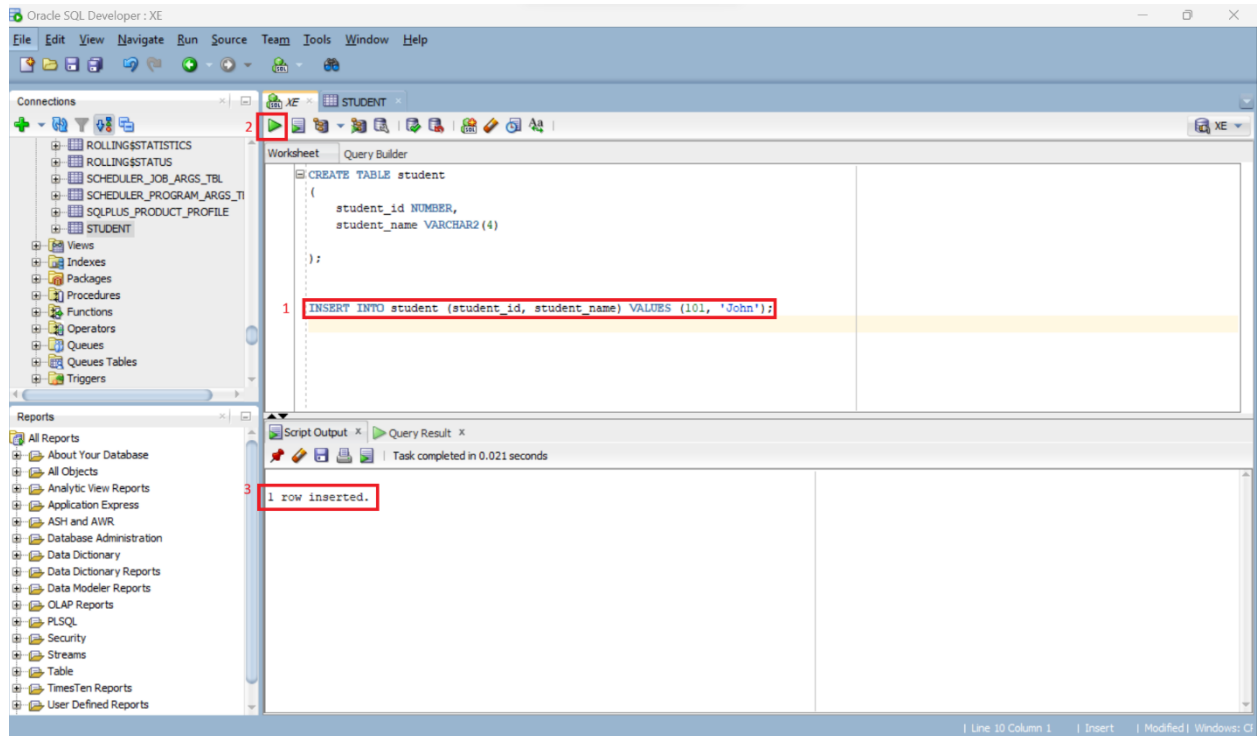
Explanation of above code -

- 1) **CREATE TABLE student** → Creates a new table called student.
- 2) **student_id NUMBER** → Adds a column student_id to store numbers.
- 3) **student_name VARCHAR2(4)** → Adds a column student_name to store text, up to 4 characters.
- 4) **(...)** → Parentheses contain the list of columns.
- 5) **;** → Ends the SQL command.

Verify table is created successfully in SQL Developer Tool, just expand XE on left panel > expand Tables > student table will be shown under that.

Step2) Insert row in table

- 1) Write the code
- 2) Click on Run button
- 3) Output will be displayed “1 row inserted”



Step3) Fetch/Display the table details

- 1) Write the code
- 2) Click on Run button
- 3) Output will be displayed with the table

